

Seismic sources

From last time

Linearised equations of motion:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = 0$$

subject to boundary conditions:

$$n_j A_{ijkl} \frac{\partial u_k}{\partial x_l} = 0.$$

Along with initial conditions on the displacement vector and its time-derivative

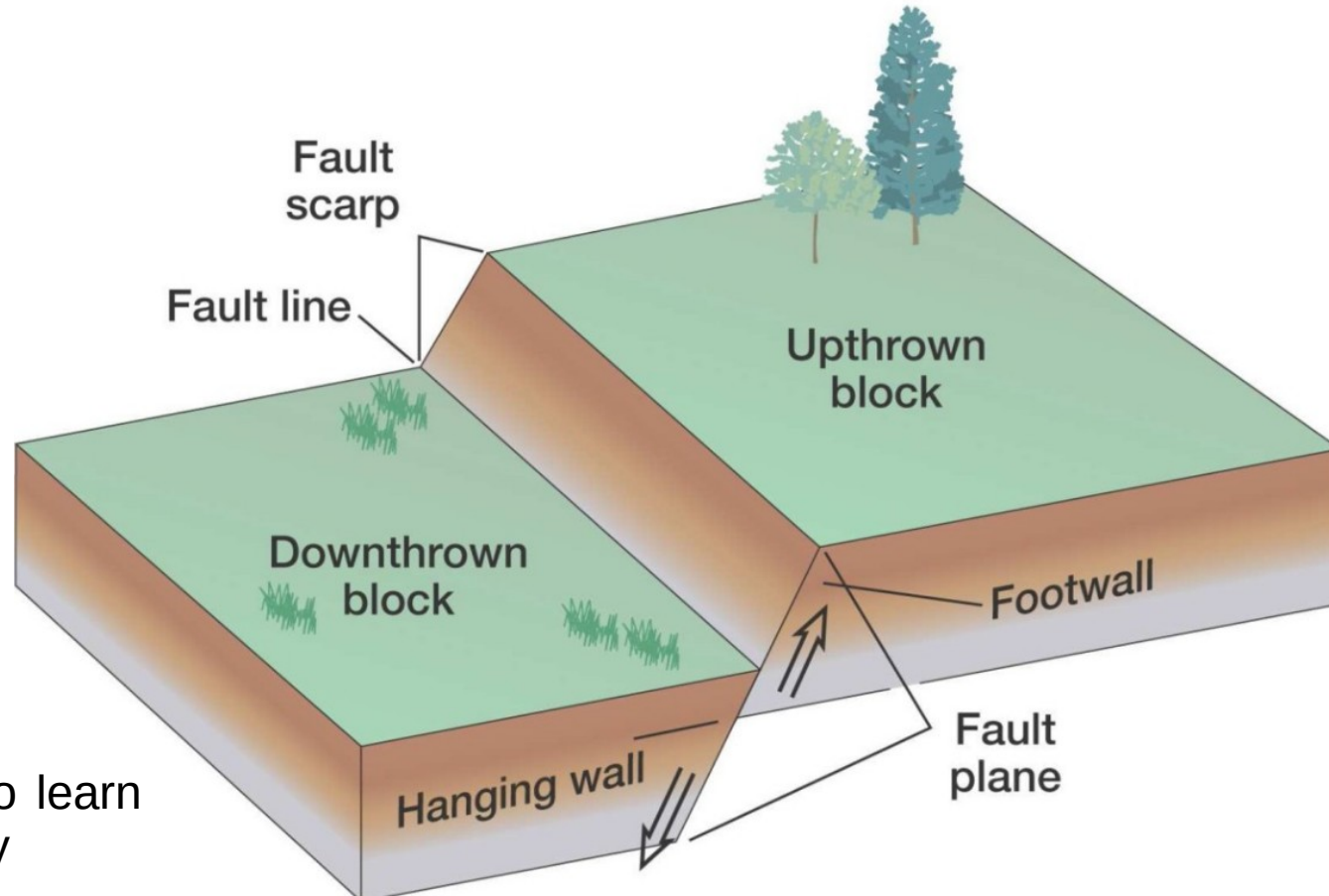
But where do earthquakes enter into the theory?

Earthquakes and tectonics



The San Andreas Fault in the Carrizo Plain of California

Idealised fault planes

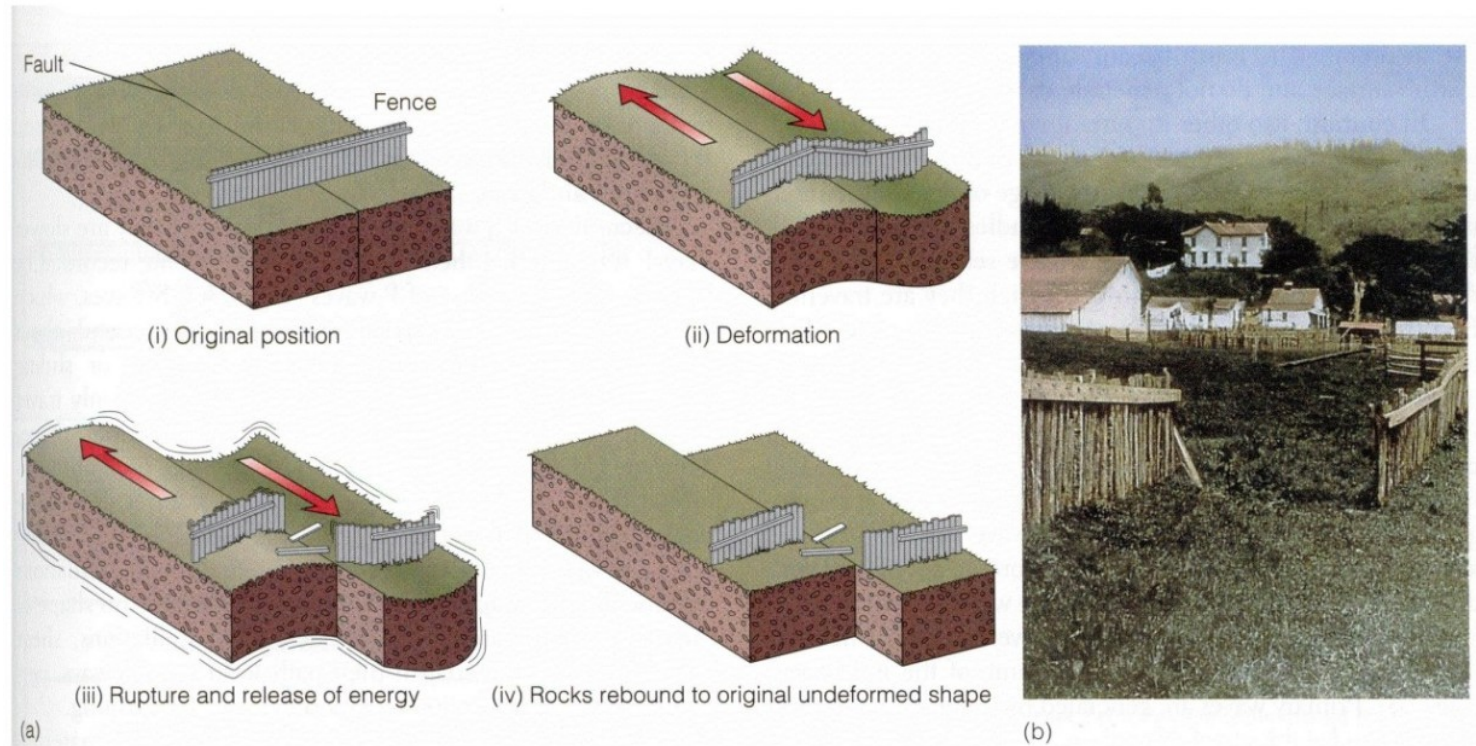


You **don't** need to learn
all this terminology

San Francisco 1906



Elastic rebound theory



Harry Fielding Reid (1859—1944)

Kinematic source approximation

Conservation of energy

Lagrangian density for finite elasticity:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, \mathbf{F}),$$

Motion of the body satisfies conservation of energy:

$$\frac{d}{dt} \int_M E(\mathbf{x}, t) d^3\mathbf{x} = 0,$$

$$E(\mathbf{x}, t) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}(\mathbf{x}, t)||^2 + W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)].$$

A body in a **stable** equilibrium configuration will remain in it – no earthquakes!

Breaking the time-translation symmetry

To model an earthquake phenomenologically, we just add an explicit time-dependence to the strain energy function:

$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, t, \mathbf{F}),$$

Equations of motion are unchanged

Constitutive equation now has an explicit time-dependence:

$$T_{ij}(\mathbf{x}, t) = \frac{\partial W}{\partial F_{ij}}[\mathbf{x}, t, \mathbf{F}(\mathbf{x}, t)].$$

Stress glut

We assume that the time-dependent strain energy function can be written:

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij}$$

The corresponding first Piola-Kirchhoff stress tensor is

$$T_{ij}(\mathbf{x}, t) = F_{ik}(\mathbf{x}, t) \left\{ 2 \frac{\partial U}{\partial C_{kj}} [\mathbf{x}, \mathbf{C}(\mathbf{x}, t)] - \bar{S}_{kj}(\mathbf{x}, t) \right\}$$

First term on right hand side is the stress predicted by elasticity, but this differs from the true stress due to the the second term

Stress glut

The **stress glut** is defined by:

$$\begin{aligned}\bar{T}_{ij}(\mathbf{x}, t) &= F_{ik}(\mathbf{x}, t) \bar{S}_{kj}(\mathbf{x}, t) \\ &= 2F_{ik}(\mathbf{x}, t) \frac{\partial U}{\partial C_{kj}}[\mathbf{x}, \mathbf{C}(\mathbf{x}, t)] - T_{ij}(\mathbf{x}, t).\end{aligned}$$

This term quantifies the failure of elasticity to correctly predict stresses in the body

By definition, the stress glut is **non-zero** within the **source region**

In the full problem, the stress glut would be determined dynamically

Under the kinematic source approximation, the stress glut is reduced to a parameter

Linearised equations of motion

We add a stress glut as a perturbation to a stress-free equilibrium configuration:

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - s \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij},$$

Linearisation proceeds almost identically to yield:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j},$$

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

Seismic source associated with an **effective body force** and **effective traction** (force per unit area) on the boundary

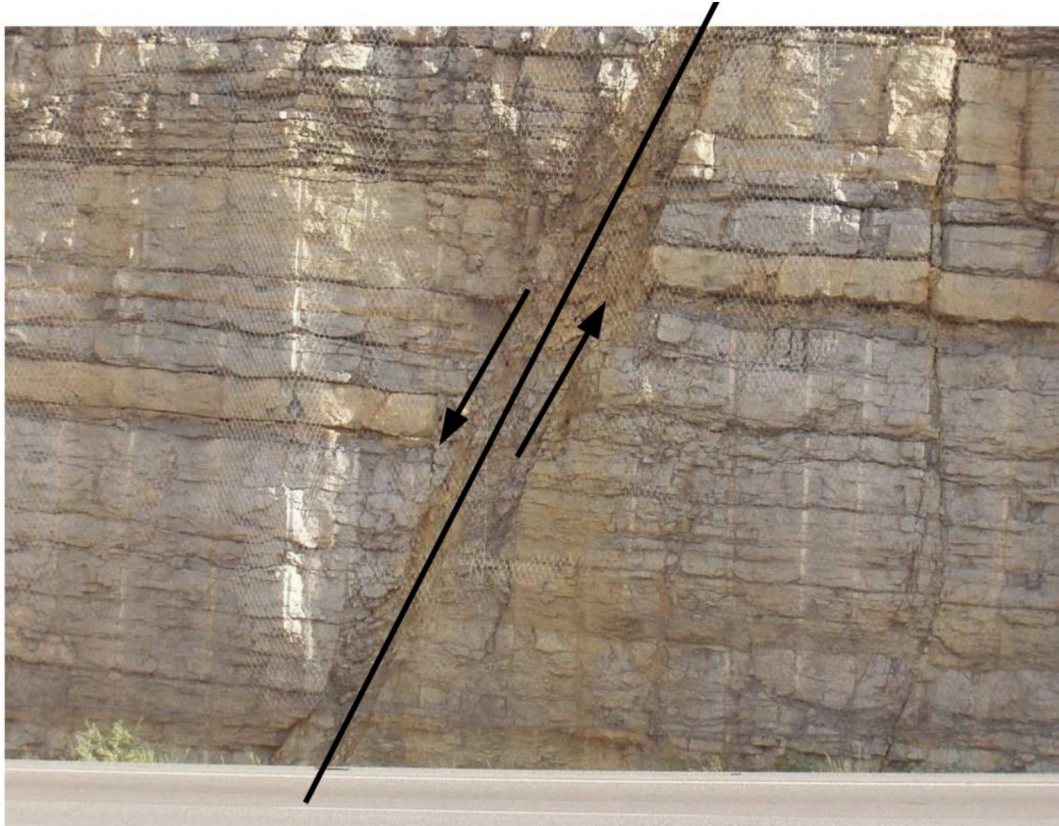
Model for an ideal fault

Real and ideal faults



Real faults involve strong non-elastic deformation across a **finite fault zone**

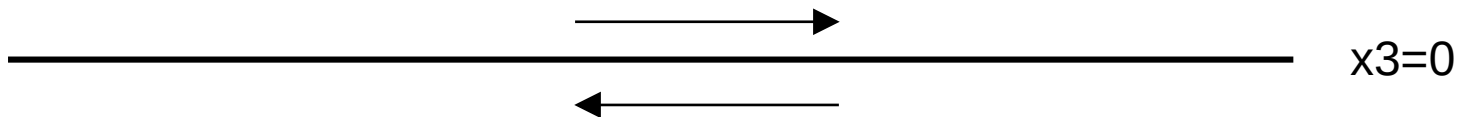
Real and ideal faults



For an **ideal fault** we collapse the fault zone to a surface across which **displacement is discontinuous**

Stress glut for an ideal fault

We consider a **planar fault** and adapt our coordinates suitably



Displacement takes the form:

$$u_i(x_1, x_2, x_3, t) = u_i^0(x_1, x_2, x_3, t) + w_i(x_1, x_2, t)H(x_3),$$

The **displacement discontinuity vector**, w , is tangential to the fault plane

Stress glut for an ideal fault

Spatial derivative of the displacement vector:

$$\frac{\partial u_i}{\partial x_j} = \frac{\partial u_i^0}{\partial x_j} + \frac{\partial w_i}{\partial x_j} H(x_3) + w_i \delta_{j3} \delta(x_3),$$

Resulting linearised stress predicted by elasticity:

$$A_{ijkl} \frac{\partial u_k^0}{\partial x_l} + A_{ijkl} \frac{\partial w_k}{\partial x_l} H(x_3) + A_{ijk3} w_k \delta(x_3).$$

Motivates the **definition** of the stress glut:

$$\bar{S}_{ij} = A_{ijk3} w_k \delta(x_3).$$

Simple generalisation to non-planar faults

Equations of motion

Equations of motion with an idealised fault:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j},$$

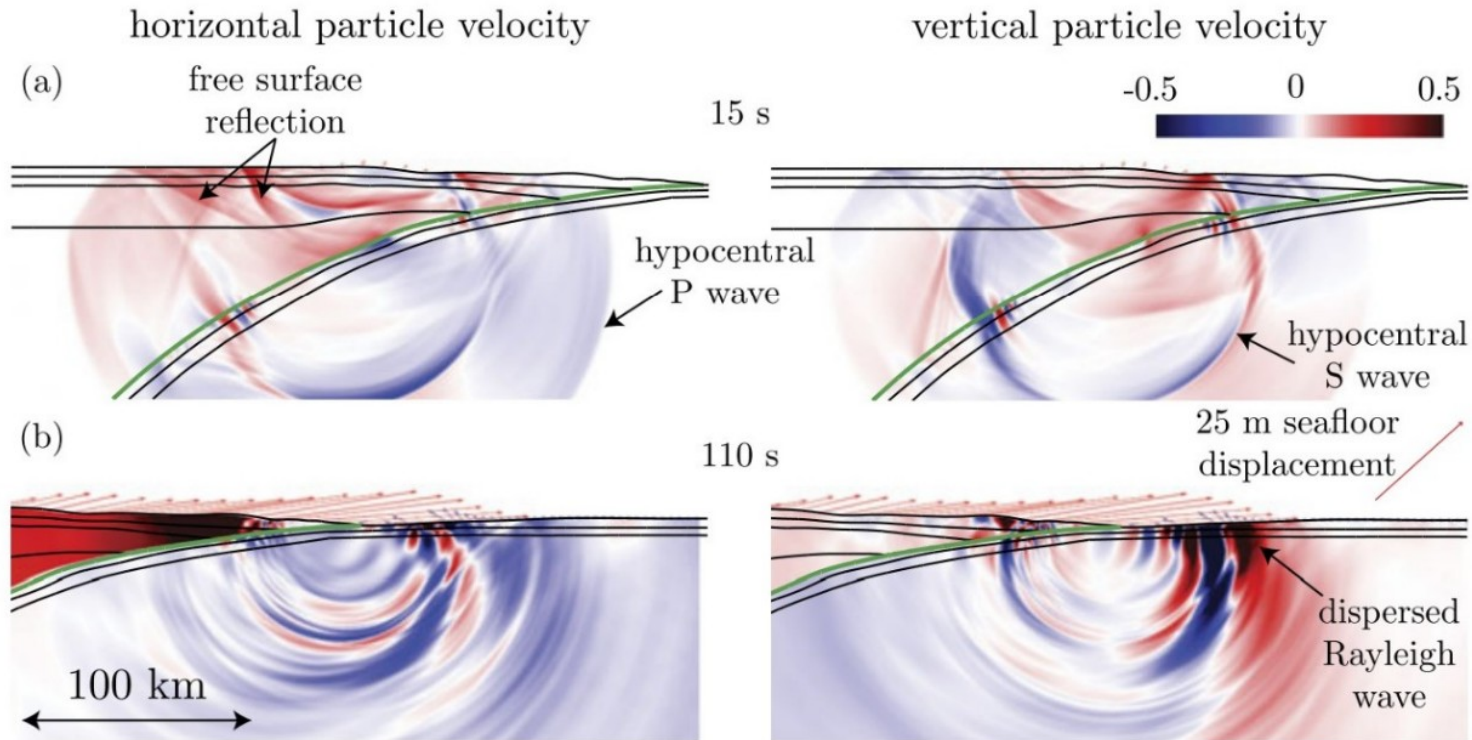
$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

$$\bar{S}_{ij} = A_{ijkl} w_k n_l \delta_{\Sigma}.$$

Parameters in these equations:

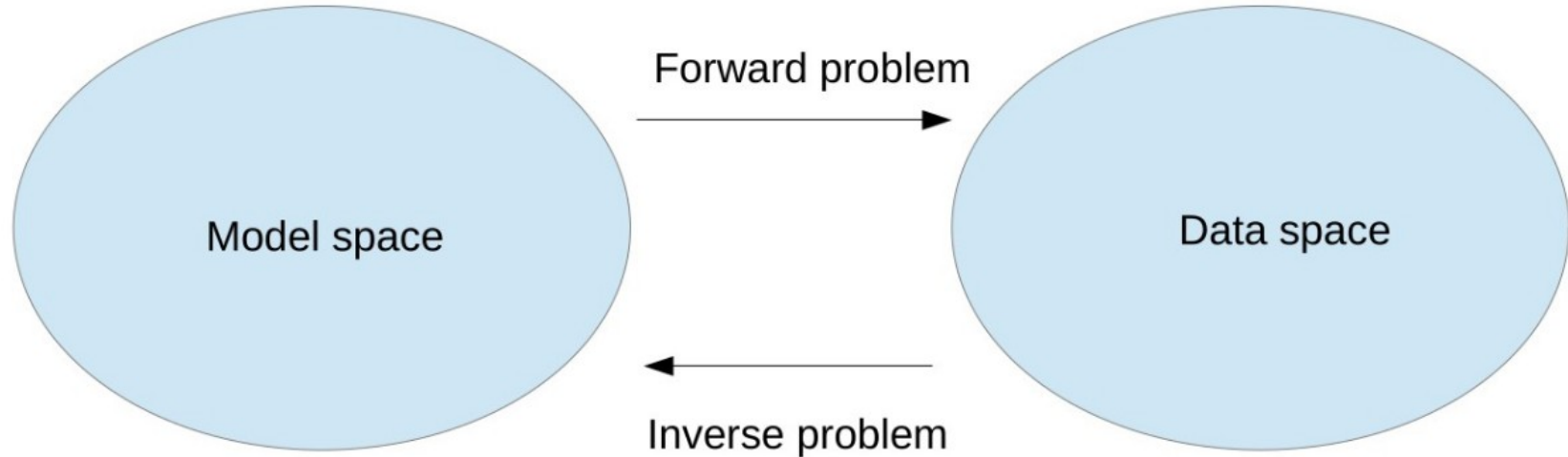
- Geometry of the Earth model
- Density and the elastic tensor
- Geometry of the fault plane and the displacement discontinuity vector

Numerical simulations



Inverse problems

Common structure



This graphical representation suggests a **symmetry** between forward and inverse problems – **this is not the case!**

Well-posed and ill-posed problems

A mathematical problem can be viewed as mapping between inputs and outputs

Such a problem is said to be **well-posed** if:

- 1) For each input there **exists** an output
- 2) For each input there exists a **unique** output
- 3) The output depends **continuously** on the input

A problem that is not well-posed is said to be **ill-posed**

The equations of mathematical physics are well-posed (or thought/hoped to be)

Inverse problems are almost always ill-posed